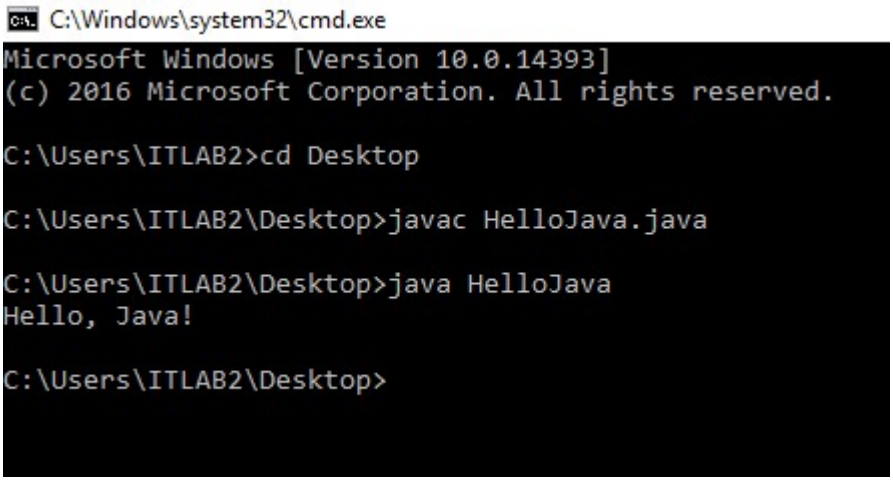
JAVA WEEK-3

1) **Write a Java program to display "Hello, Java!" and demonstrate the basic structure of a Java program.**

```
class HelloJava {

    public static void main(String[] args) {

        System.out.println("Hello, Java!");
    }
}
```



2) **Write a Java program to declare variables of different primitive data types (`byte`, `short`, `int`, `long`, `float`, `double`, `char`, and `boolean`) and display their values.**

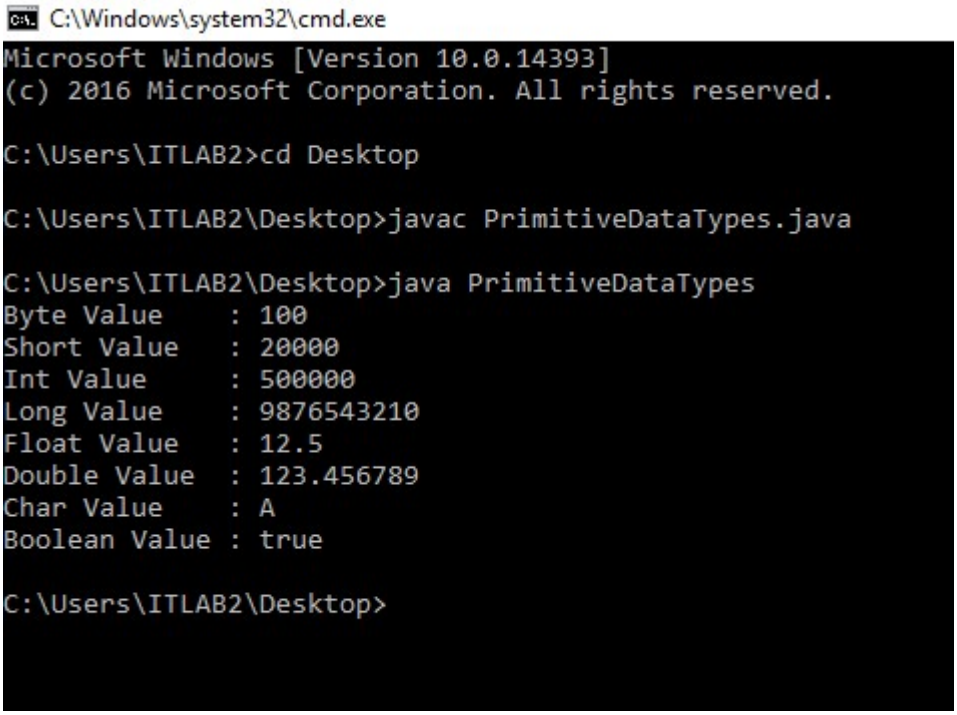
```
class PrimitiveDataTypes {
    public static void main(String[] args) {

        byte BV = 100;
```

```
        short SV = 20000;
        int IV = 500000;
        long LV = 9876543210L;
        float FV = 12.5f;
        double DV = 123.456789;
        char CV = 'A';
        boolean BOLV = true;



        System.out.println("Byte Value    : " + BV);
        System.out.println("Short Value   : " + SV);
        System.out.println("Int Value     : " + IV);
        System.out.println("Long Value    : " + LV
        System.out.println("Float Value   : " + FV);
        System.out.println("Double Value  : " + DV);
        System.out.println("Char Value    : " + CV);
        System.out.println("Boolean Value : " + BOLV);
    }
}
```



3) **Write a Java program to perform arithmetic operations (addition, subtraction, multiplication, division, and modulus) on two integer variables.**

```
class ArithmeticOperations {

  public static void main(String[] args) {


    int a = 20;

    int b = 6;


    int addition = a + b;

    int subtraction = a - b;

    int multiplication = a * b;

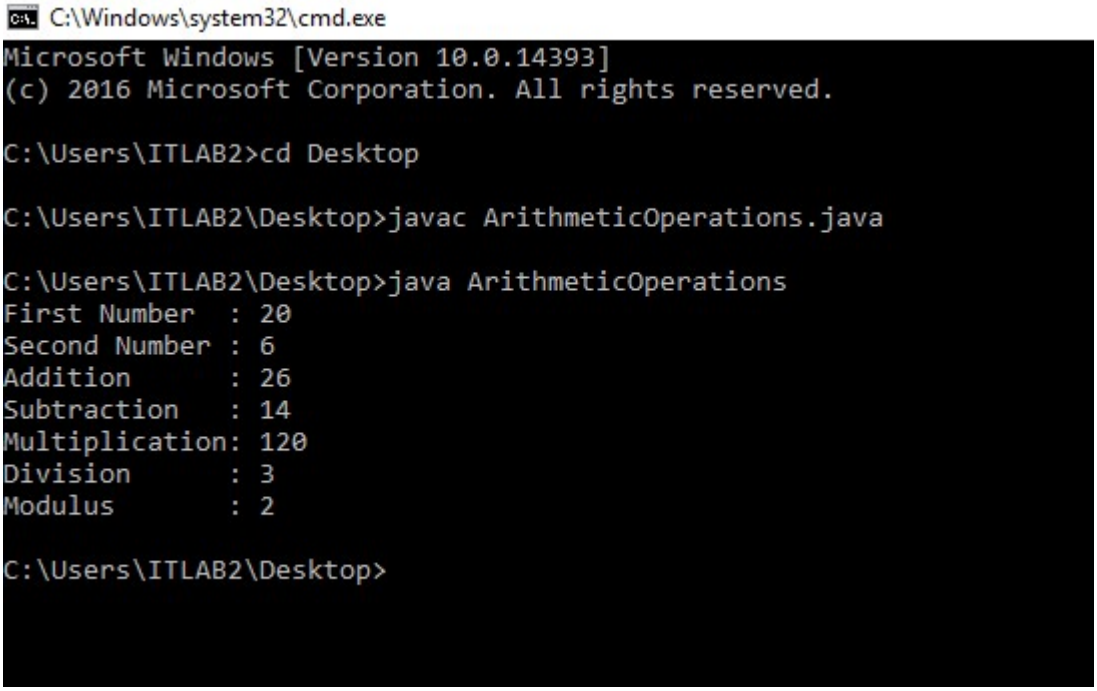
    int division = a /b;

    int modulus =a % b;



    System.out.println("First Number  : " +a);
```

```
        System.out.println("Second Number : " + b);
        System.out.println("Addition      : " + addition);
        System.out.println("Subtraction   : " + subtraction);
        System.out.println("Multiplication: " + multiplication);
        System.out.println("Division      : " + division);
        System.out.println("Modulus       : " + modulus);
    }
}
```



4) **Write a Java program to perform arithmetic operations on floating-point numbers and display the results.**

```
class FloatingPointArithmetic {
    public static void main(String[] args) {
```

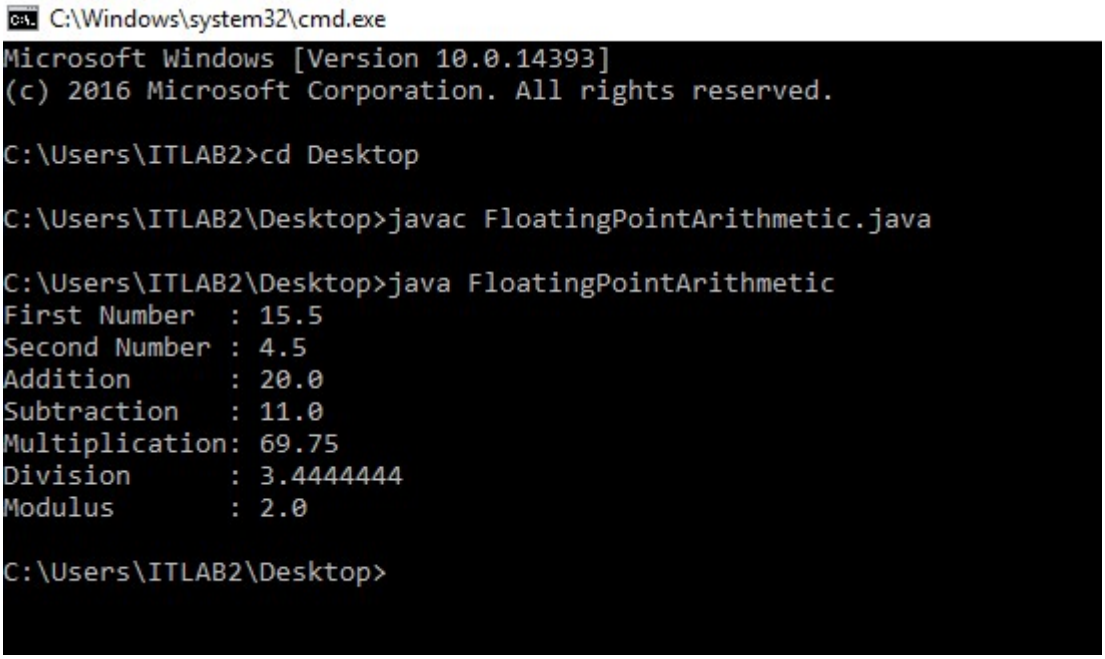
```
        float a = 15.5f;
        float b = 4.5f;



        float addition = a + b;
        float subtraction = a - b;
        float multiplication = a * b;
        float division =a/ b;
        float modulus = a % b;



        System.out.println("First Number  : " + a);
        System.out.println("Second Number : " + b);
        System.out.println("Addition      : " + addition);
        System.out.println("Subtraction   : " + subtraction);
        System.out.println("Multiplication: " + multiplication);
        System.out.println("Division      : " + division);
        System.out.println("Modulus       : " + modulus);
    }
}
```



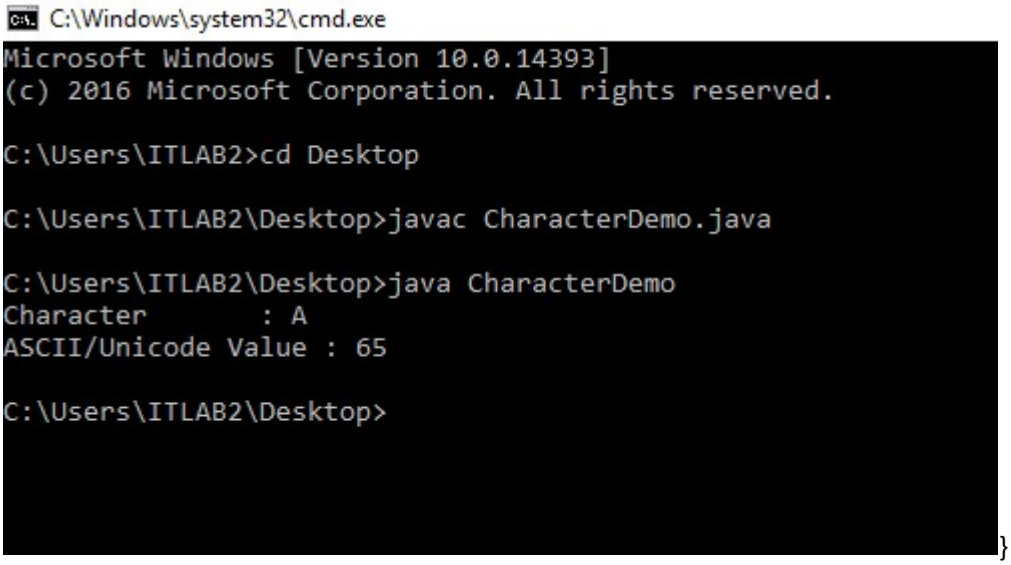
5) **Write a Java program to demonstrate character data type by displaying a character and its corresponding ASCII/Unicode value.**

```
class CharacterDemo {
   public static void main(String[] args) {

      char ch = 'A';

      int asciiValue = ch;

      System.out.println("Character      : " + ch);
      System.out.println("ASCII/Unicode Value : " + asciiValue);
   }
```



}

6) **Write a Java program to demonstrate the use of Boolean variables and Boolean expressions.**

```
class BooleanDemo {
    public static void main(String[] args) {

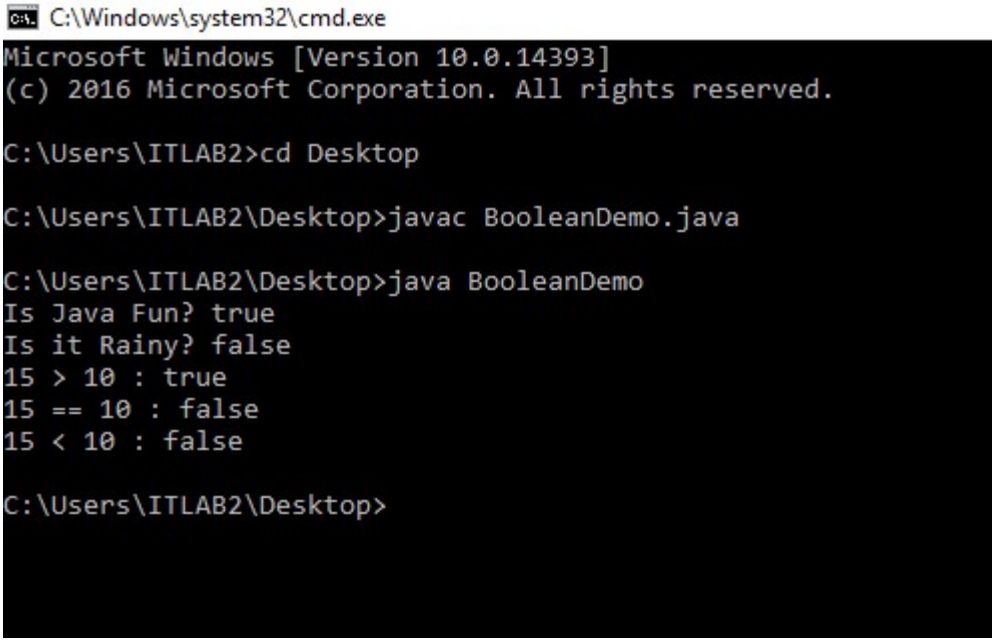
        boolean isJavaFun = true;
        boolean isRainy = false;
        System.out.println("Is Java Fun? " + isJavaFun);
        System.out.println("Is it Rainy? " + isRainy);



        int num1 = 15;
        int num2 = 10;

        boolean greaterThan = num1 > num2;
        boolean equalTo = num1 == num2;
        boolean lessThan = num1 < num2;
```

```
        System.out.println(num1 + " > " + num2 + " : " + greaterThan);
        System.out.println(num1 + " == " + num2 + " : " + equalTo);
        System.out.println(num1 + " < " + num2 + " : " + lessThan);
    }
}
```



7) **Write a Java program to declare and initialize different types of variables and display their values.**

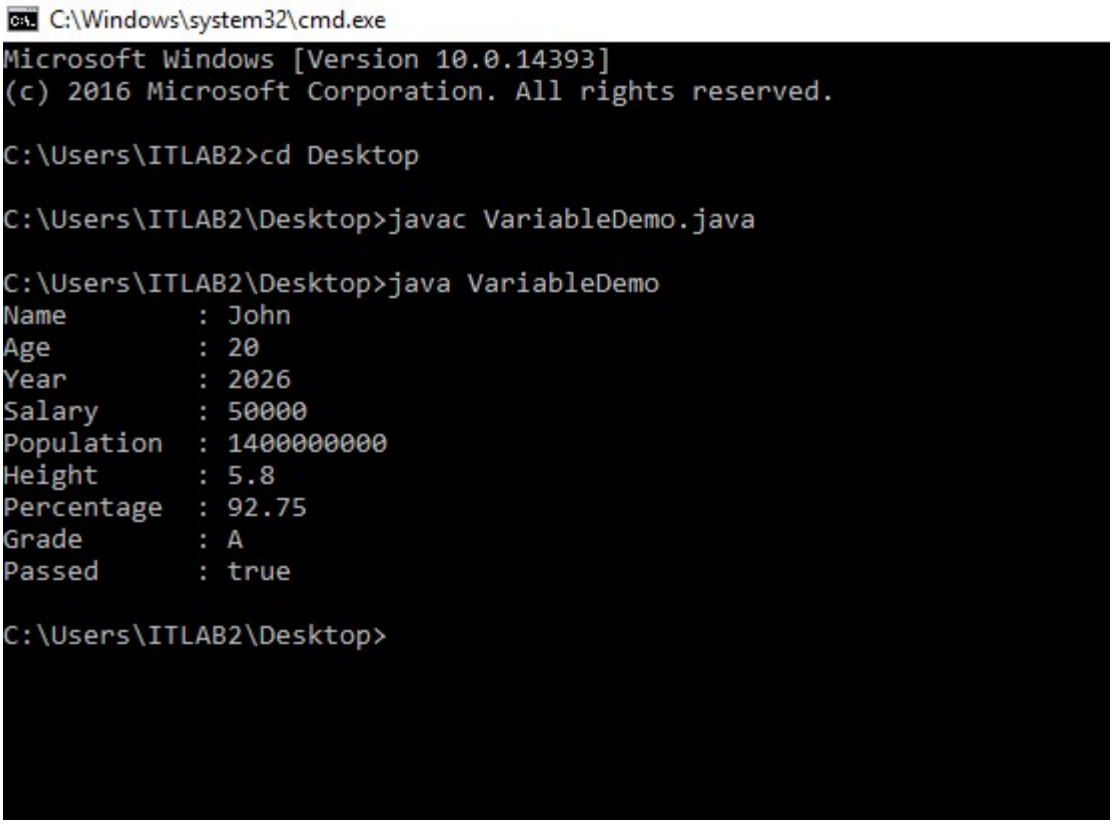
```
class VariableDemo {
    public static void main(String[] args) {

        byte age = 20;
        short year = 2026;
        int salary = 50000;
        long population = 1400000000L;
        float height = 5.8f;
        double percentage = 92.75;
```

```
        char grade = 'A';
        boolean isPassed = true;
        String name = "John";



        System.out.println("Name       : " + name);
        System.out.println("Age        : " + age);
        System.out.println("Year       : " + year);
        System.out.println("Salary     : " + salary);
        System.out.println("Population  : " + population);
        System.out.println("Height     : " + height);
        System.out.println("Percentage  : " + percentage);
        System.out.println("Grade      : " + grade);
        System.out.println("Passed     : " + isPassed);
    }
}
```



8) **Write a Java program to swap the values of two variables using a temporary variable.**

```
class Swap {
   public static void main(String[] args) {

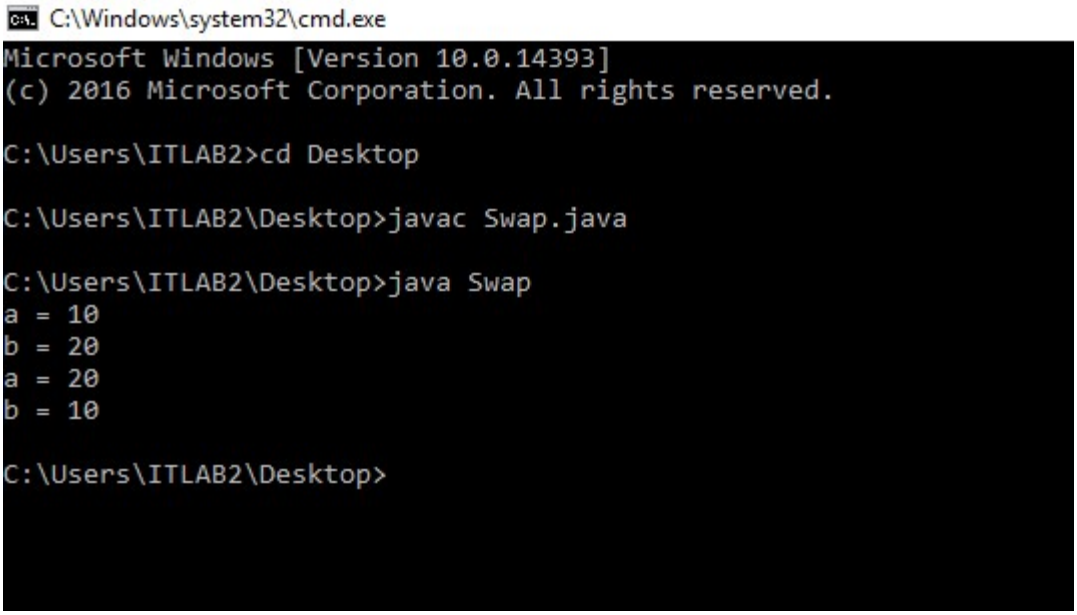
      int a = 10;
      int b = 20;
      int temp;


      System.out.println("a = " + a);
      System.out.println("b = " + b);

      temp = a;
```

```
        a = b;
        b = temp;

        System.out.println("a = " + a);
        System.out.println("b = " + b);
    }
}
```



9) **Write a Java program to demonstrate automatic (widening) type conversion between primitive data types.**

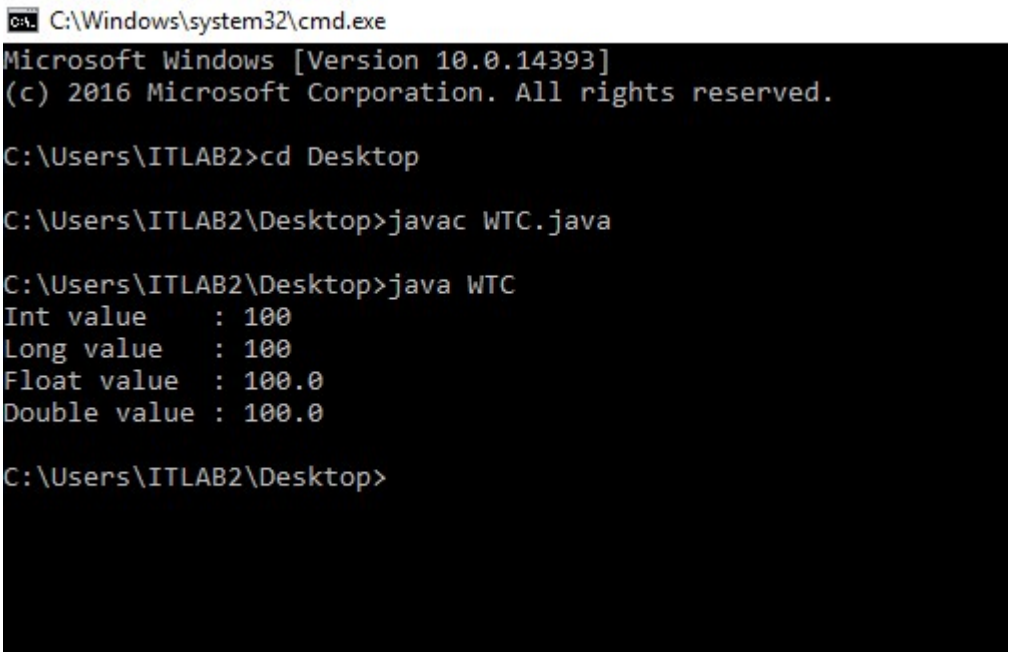
```
class WTC {
    public static void main(String[] args) {
```

```
        int intValue = 100;



        long longValue = intValue;
        float floatValue = intValue;
        double doubleValue = intValue;


        System.out.println("Int value    : " + intValue);
        System.out.println("Long value   : " + longValue);
        System.out.println("Float value  : " + floatValue);
        System.out.println("Double value : " + doubleValue);
    }
}
```



10) **Write a Java program to demonstrate explicit (narrowing) type casting from one primitive data type to another.**

```
class NTC {
```

```
    public static void main(String[] args) {



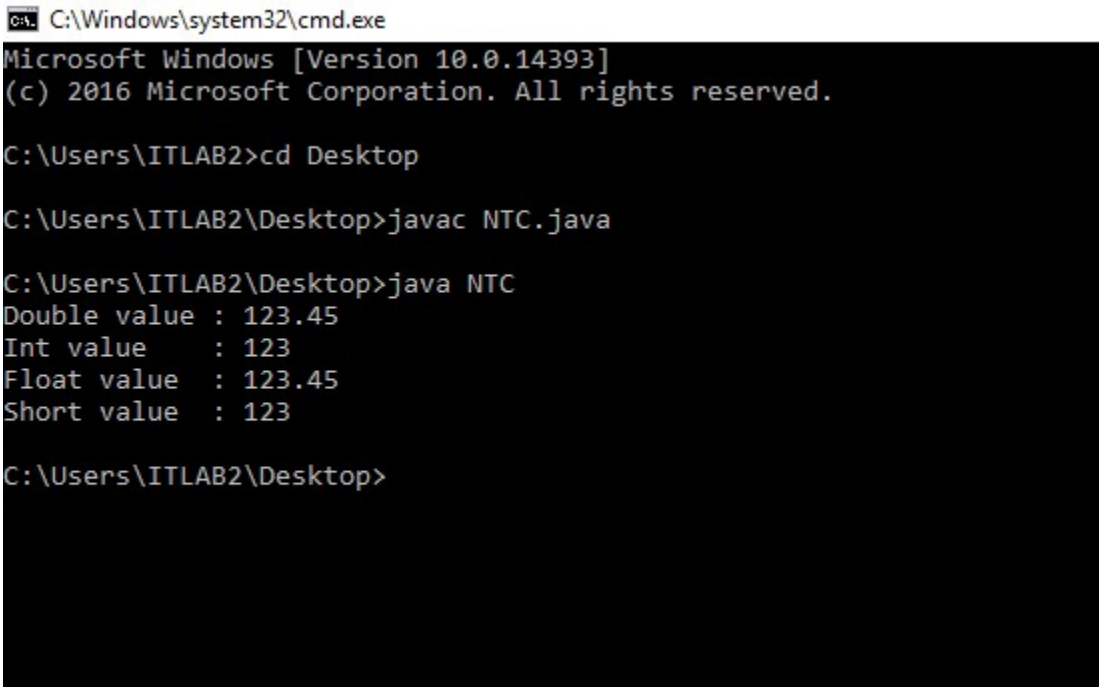
        double doubleValue = 123.45;



        int intValue = (int) doubleValue;
        float floatValue = (float) doubleValue;
        short shortValue = (short) doubleValue;



        System.out.println("Double value : " + doubleValue);
        System.out.println("Int value    : " + intValue);
        System.out.println("Float value  : " + floatValue);
        System.out.println("Short value  : " + shortValue);
    }
}
```



11) **Write a Java program to convert a character to its ASCII/Unicode value and vice versa using type casting.**

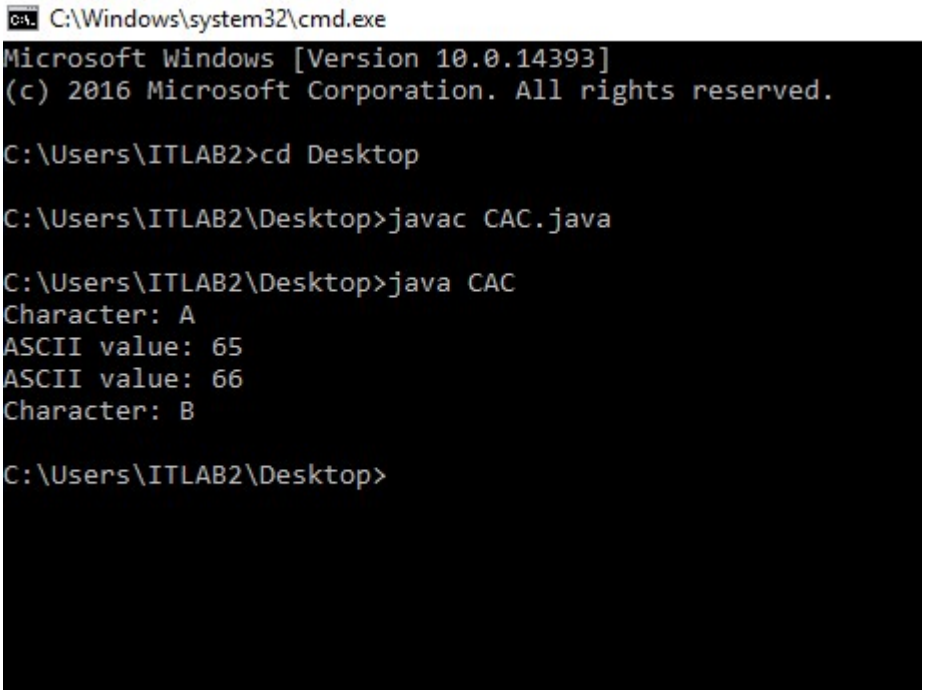
```
class CAC {
   public static void main(String[] args) {


      char ch = 'A';
      int ascii = ch;  // implicit casting


      int value = 66;
      char ch2 = (char) value; // explicit casting
```

```
        System.out.println("Character: " + ch);
        System.out.println("ASCII value: " + ascii);

        System.out.println("ASCII value: " + value);
        System.out.println("Character: " + ch2);
    }
}
```



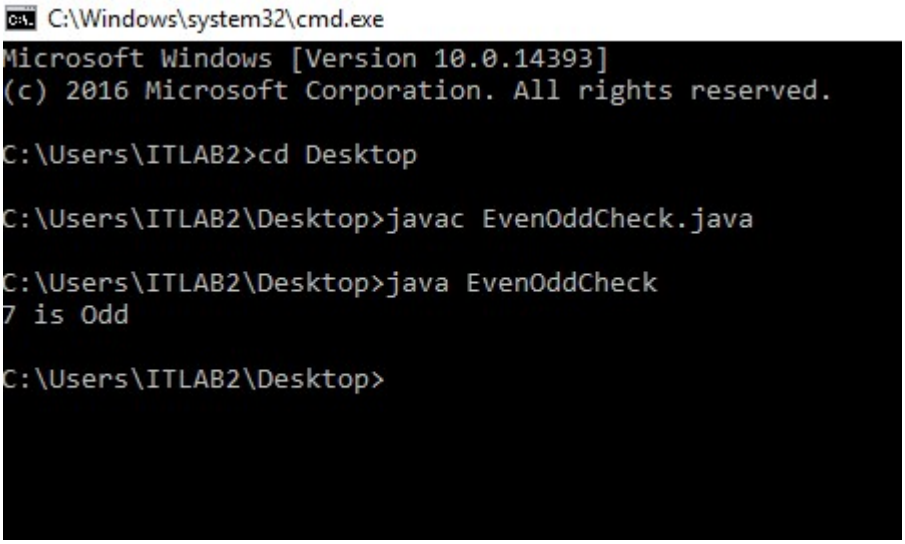
12) **Write a Java program to check whether a given number is even or odd using the `if-else` control statement.**

```
class EvenOddCheck {
    public static void main(String[] args) {

        int number = 7;

        if (number % 2 == 0) {
```

```
        System.out.println(number + " is Even");
    } else {
        System.out.println(number + " is Odd");
    }
  }
}
```



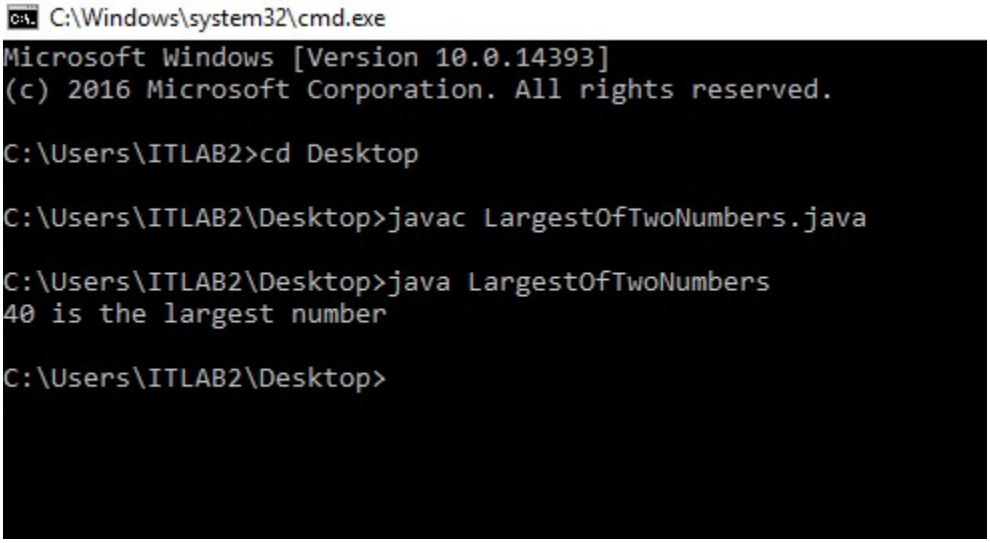
13) Write a Java program to find the largest of two numbers using the `if-else` statement.

```
class LargestOfTwoNumbers {
  public static void main(String[] args) {

    int a = 25;
    int b = 40;

    if (a > b) {
      System.out.println(a + " is the largest number");
    } else {
```

```
        System.out.println(b + " is the largest number");
      }
   }
}
```



14) **Write a Java program to demonstrate the use of Java reserved keywords by creating valid identifiers and explaining why keywords cannot be used as variable names.**

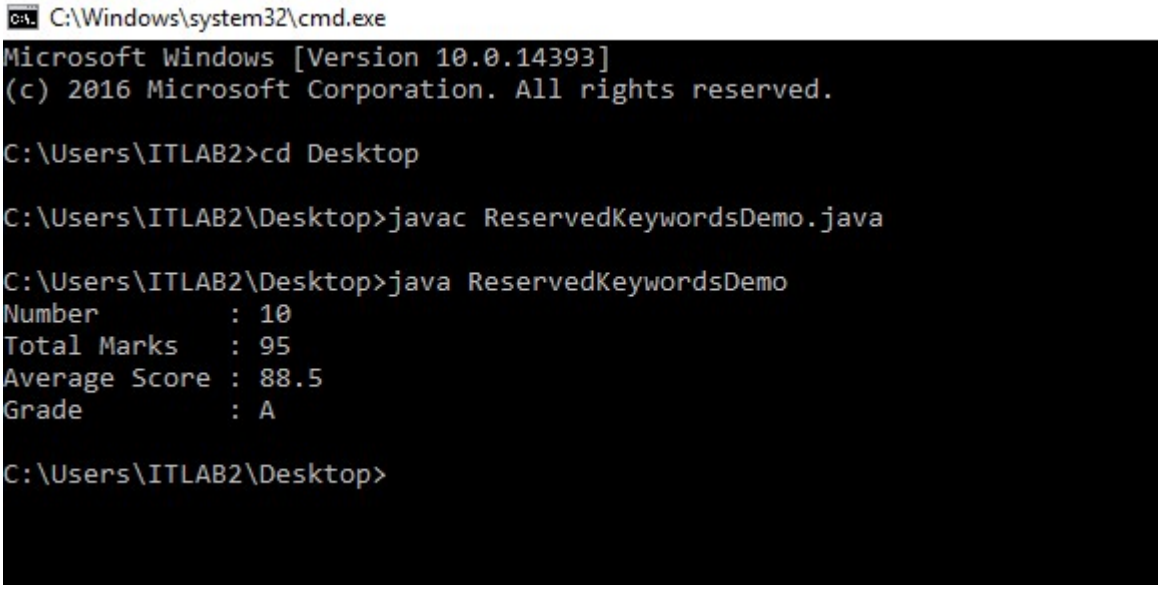
```
class ReservedKeywordsDemo {
   public static void main(String[] args) {


      int number = 10;
      int totalMarks = 95;
      float average_score = 88.5f;
      char grade = 'A';


      System.out.println("Number       : " + number);
```

```
        System.out.println("Total Marks   : " + totalMarks);
        System.out.println("Average Score : " + average_score);
        System.out.println("Grade         : " + grade);

        /*
         * Keywords like int, class, public, static, etc.
         * cannot be used as variable names because they are
         * reserved by Java for predefined purposes.
         *
         * Example (INVALID - will cause error):
         * int int = 5;
         * class class = 10;
         */
    }
}
```



15) **Write a Java program to demonstrate the use of the for loop by displaying numbers from 1 to 10.**

```
class ForLoopDemo {
```

```
    public static void main(String[] args) {


        for (int i = 1; i <= 10; i++) {
            System.out.println(i);
        }
    }
}
```

